

2. Choose your Thread border router strategy
3. Add 1–3 Matter devices + sanity-check local control
4. Create one “proof” automation
5. Lock down remote access and automate backups

1) Pick your hardware: the three realistic paths

For a weekend build, the win condition is simple: use Home Assistant OS. Home Assistant’s Raspberry Pi guide calls Home Assistant OS the recommended installation type for most users, because it bundles Core + Supervisor and supports add-ons (without you managing the underlying OS). (Also worth noting: Home Assistant has recommended migrating Supervised users to Home Assistant OS for the full, supported experience.)

Option A: Home Assistant Green (fastest success)

Home Assistant Green is positioned as the easiest way to start – plug-and-play, already installed.

Pick this if: you want smart-home results, not a hardware weekend.

Option B: Raspberry Pi (classic DIY)

Home Assistant’s Pi guide walks you through flashing Home Assistant OS with Raspberry Pi Imager and accessing it at `home-assistant.local:8123`.

Pick this if: you already own a Pi and want a compact hub.

Option C: Small x86 mini PC (most “set and forget”)

A NUC-style mini PC with an SSD is boring in the best way: fast storage, low maintenance, room for add-ons. The Generic x86-64 install guide highlights the BIOS/UEFI prerequisites (UEFI on, Secure Boot off).

Pick this if: you expect your setup to grow beyond “a few lights”.

Fast Track advice for India: If you’re buying hardware in 2026, Green or an x86 mini PC is usually the cleanest long-term base. Pi works well too – just don’t treat a cheap microSD card like enterprise storage (and don’t hide the hub inside a metal cabinet).

2) Install Home Assistant OS (step-by-step)

Raspberry Pi install (the “no mistakes” checklist)

Follow the official flow; these steps are straight from Home Assistant’s Raspberry Pi installation guide.

You need: Pi 3/4/5, power supply, microSD (or SSD if supported), and Ethernet for first boot.

1. Install Raspberry Pi Imager.
2. Imager → Choose OS → Other specific-purpose OS → Home assistants and home automation → Home Assistant.
3. Choose the correct image for your Pi model, pick storage, write.

4. Insert storage into Pi, connect Ethernet, power on.
5. On your computer, open `homeassistant.local:8123` (or use the Pi’s IP address if your network is strict).
6. Once the UI loads, move to onboarding.

Practical tip: Do the first boot on Ethernet. Wi-Fi can come later; stability first.

x86 mini PC install (quick reality check)

If you’re using x86 hardware, confirm UEFI boot mode and Secure Boot disabled as the guide notes. Then follow the official installer flow for your target device.

3) Onboarding: do it properly (or automations will lie)

Home Assistant uses your home location to set time zone, unit system, currency, and to create a home zone used in presence-based automations. This matters because “sunset”, “weekday”, and “I’m home” are the three most common automation ingredients – and they all depend on correct basics.

Step-by-step onboarding

1. Create your admin account (strong password; treat this like your router admin login).
2. Set your home location precisely; Home Assistant uses it to configure the time zone and define the home zone (default radius ~100 m).
3. If anything looks off later, Home Assistant documents changing these basics in General settings.
4. Let Home Assistant auto-discover devices on your network, but don’t accept every prompt blindly.

3.1 Create Areas first (10 minutes now, hours saved later)

Before you add devices, go to Settings → Areas and create the rooms you actually use: Living Room, Bedroom, Kitchen, Study, Balcony, Entrance.

Then follow one rule: every device gets an Area assignment the moment it’s added.

3.2 A naming system that scales

Use a simple, consistent pattern:

- Prefix by room: LR / BR / KT / EN
- Describe function: “Ceiling”, “Lamp”, “Plug”, “Sensor”
Example: “LR Ceiling Light”, “KT Plug – Microwave”, “EN Door Sensor”.

This seems fussy – until you have 40 entities and your automation editor becomes a search box.

4) Add-ons: why Home Assistant OS feels like cheating

Add-ons are what turn Home Assistant from “a dashboard” into